Homework 10 - 10/13 Content

MATH 1271, Garrett Credi

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Problems listed as AC are found in the Active Calculus textbook at the end of the listed topic. You should be finding these in the *Exercises* section of the page.

Problems listed as APEX are found in the APEX Calculus Textbook again at the end of the listed topic. You should be finding these in the *Exercises* section of the page.

Problems listed as PAULS-PRACT are found in Paul's Online Math Notes - Review Problems while problems listed as PAULS-ASSIGN are found in Paul's Online Math Notes - Assignment Problems and are indexed via [Topic]-[Section]-[Problem Number]

1 Type 1 - Product Rule and Quotient Rule

Choose 6 of the following problems to solve.

1. AC 2.3.1-15 (each of these counts as one problem)
2. APEX 2.4.15-36 (each of these counts as one problem)
3. APEX 2.4.41-44 (each of these counts as one problem)
4. PAULS-ASSIGN.Derivatives.Product and Quotient Rule.1-19 (each of these counts as one problem)

2 Type 2 - Trig Derivatives

Choose 6 of the following problems to solve.

1. AC 2.2.1-5 (each of these counts as one problem)
2. PAULS-ASSIGN.Derivatives.Derivatives of Trig Functions.1-28 EXCEPT 25 (each of these counts as one problem)

3 Type 3 - Challenge Problem

The product rule

$$\frac{d}{dx}(f(x)g(x)) = (\frac{d}{dx}f(x)) \cdot g(x) + f(x) \cdot (\frac{d}{dx}g(x)) \tag{1}$$

is so important that it gets *another* name - it's alternatively called the Leibniz Rule. Demonstrate why the product rule is so important by proving

1. The product rule forces $\frac{d}{dx}(1) = 0$. (Think of $\frac{d}{dx}1^2$)
2. The product rule forces the constant multiple rule $\frac{d}{dx}(c \cdot f(x)) = c\frac{d}{dx}(f(x))$.
3. Together with $\frac{d}{dx}x = 1$, the product rule forces the power rule $\frac{d}{dx}x^n = nx^{n-1}$. (Hint $x^n = x \cdot x^{n-1}$ for $n \geq 1$).

I.e. for each of the above goals, use only the product rule (plus the additional assumption for the third one) to prove that the forced rule must hold.